

Practice Questions for Final Exam

Stat 332: Sampling and Experimental Design (Winter 2026)

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Notes:

- All of the following questions are from past STAT 332 midterm exams, when I taught the class.
 - However, they are not from a single midterm, and thus are not a good indication of the length of your upcoming midterm.
 - In the interest of space, I'm not including all the formulas I provided with the questions, as you can simply look them up in the slides.
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Question #1:

This question is concerned with sums of squares and their expectations in a two-way ANOVA.

(a) Using that

$$SSA + SSB + SSAB = SSTR = n_0 \sum_{i=1}^I \sum_{j=1}^J (\bar{y}_{ij\bullet} - \bar{y}_{\bullet\bullet})^2$$

show that

$$SSTO = SSAB + SSA + SSB + SSE$$

(b) Let $MSTR = SSTR/(IJ - 1)$; compute $E(MSTR)$.

Question #2:

This question is concerned with providing more insight into least square estimation in the two-way Anova model and into the Tukey Test for additivity.

- (a) Consider a two-way Anova model with a single observation per cell (i.e., $n_0 = 1$ in our standard two-way Anova model); that is,

$$Y_{ij} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \varepsilon_{ij}$$

where ε_{ij} i.i.d. $N(0, \sigma^2)$ and where

$$\sum_{i=1}^I \alpha_i = 0 \quad \sum_{j=1}^J \beta_j = 0 \quad \sum_{j=1}^J (\alpha\beta)_{ij} = 0 \quad \forall i\text{'s} \quad \sum_{i=1}^I (\alpha\beta)_{ij} = 0 \quad \forall j\text{'s}$$

Show that the least squares estimators of μ , α_i and β_j are given by

$$\begin{aligned}\hat{\mu} &= \bar{y}_{..} = \frac{1}{IJ} \sum_{i=1}^I \sum_{j=1}^J y_{ij} \\ \hat{\alpha}_i &= \bar{y}_{i.} - \bar{y}_{..} = \frac{1}{J} \sum_{j=1}^J y_{ij} - \bar{y}_{..} \\ \hat{\beta}_j &= \bar{y}_{.j} - \bar{y}_{..} = \frac{1}{I} \sum_{i=1}^I y_{ij} - \bar{y}_{..}\end{aligned}$$

You can assume that all 2nd derivatives are > 0 , and there is thus no need for you to compute those 2nd derivatives to confirm that the above are indeed minimums. You do not have to compute the least squares estimator of $(\alpha\beta)_{ij}$.

- (b) Using that fact that $(\widehat{\alpha\beta})_{ij} = y_{ij} - \bar{y}_{i.} - \bar{y}_{.j} + \bar{y}_{..}$, show that the fitted values simplify to

$$\hat{y}_{ij} = y_{ij}$$

- (c) Recall that the residuals are defined as $e_{ij} = y_{ij} - \hat{y}_{ij}$, and that

$$\text{SSE} = \sum_{i=1}^I \sum_{j=1}^J (y_{ij} - \hat{y}_{ij})^2 = \sum_{i=1}^I \sum_{j=1}^J e_{ij}^2$$

What is(are) the implication(s) of using the two-way Anova model given in part (a)?

- (d) If we are still interested by the interaction and only have a single observation per cell, an attractive option is to use the model behind the Tukey Test for additivity; that is,

$$Y_{ij} = \mu + \alpha_i + \beta_j + \lambda\alpha_i\beta_j + \varepsilon_{ij}$$

with ε_{ij} i.i.d. $N(0, \sigma^2)$ and the same constraints as in part (a).

Using the fact that the estimates of μ , α_i and β_j are the exact same as given in part (a) above, show that the least squares estimator of λ is given by

$$\hat{\lambda} = \frac{\sum_{i=1}^I \sum_{j=1}^J \hat{\alpha}_i \hat{\beta}_j y_{ij}}{\left(\sum_{i=1}^I \hat{\alpha}_i^2 \right) \left(\sum_{j=1}^J \hat{\beta}_j^2 \right)}$$

Question #3:

A car magazine made a small experiment to investigate if the use of a higher octane gasoline would increase miles per gallon (mpg) achieved by an automobile. Since driving habits can have an important impact on mpg, it was decided to use a randomized block design with driver as the blocking factor. The 9 observations are given in the table below.

- (a) Using the handout with the computational formulas, complete the Anova table below.
- (b) Test (at the 5% level) whether or not gasoline octane effects mpg. Note: $F_{2,4;0.95} = 6.94$.
- (c) Repeat part (b) but using a one-way ANOVA model, thus ignoring that blocking was used. Do you reach the same conclusion on the effect of gasoline octane on mpg as in part (b). Note: $F_{2,6;0.95} = 5.14$. *Note: you do not have to provide the one-way ANOVA table, but you can if it helps you.*

Driver	Octane		
	87	89	93
1	33.5	34.2	35.3
2	33.2	33.8	36.1
3	33.0	31.3	32.7

$$\text{Note: } \sum_{i=1}^I \sum_{j=1}^r y_{ij}^2 = 10217.6 \quad \text{and} \quad \sum_{i=1}^I \sum_{j=1}^r y_{ij} = 303$$

Source	d.f.	SS	MS	F value
Octane		4.8467		
Blocks				
Error		3.3467		
Total				

Question #4:

Suppose that we want to investigate the effects of time (factor A), pressure (factor B) and temperature (factor C) on the defect rate (per 1000 parts) in a given industrial process. To this end, a 2^3 factorial design. Two replicates/observations were collected for each of the 8 treatments. Those 2 observations and their means are given in the last 3 column of the table below.

(a) Complete the algebraic signs matrix below for this design.

(b) Using that data, compute the contrast/effect for factor A.

Recall that:

$$\text{effect} = \frac{1}{4} \sum_{i=1}^2 \sum_{j=1}^2 \sum_{k=1}^2 c_{ijk} \bar{y}_{ijk\bullet} \quad \text{where} \quad \bar{y}_{ijk\bullet} = \frac{1}{r} \sum_{\ell=1}^r y_{ijk\ell}$$

(c) If $\text{SSE} = 0.70$, test (at the 5% level) if the main effect of factor A is statistically significant. *Recall that:*

$$F^* = \frac{\text{MSA}}{\text{MSE}} = \frac{\text{SSA}}{\text{SSE}/(n-8)} = \frac{(n/4) \times (\text{effect}_A)^2}{\text{SSE}/(n-8)}$$

and that, we reject H_0 (in favour of H_1) if $F_{obs}^* > F_{1,n-8;1-\alpha}$, where and $F_{1,n-8;1-\alpha}$ can be found in the provided F distribution table.

A	B	C	AB	AC	BC	ABC	Defect rate	$\bar{y}_{ijk\bullet}$
—			+	+			10.3 9.7	10.00
+			—	—			9.9 10.4	10.15
—			—	+			11.9 11.8	11.85
+			+	—			12.1 12.5	12.30
—			+	—			15.0 15.5	15.25
+			—	+			15.6 16.0	15.80
—			—	—			17.2 16.8	17.00
+			+	+			17.0 17.2	17.10

Question #5:

This question is from the final when I taught STAT 332 online during the pandemic. Since all exams were take-home exams, this question looks a little different as the students were asked to use R to complete it. For that reason (and because there is likely not enough time), it will not be covered during the Zoom review session. However, I will pose the solutions, as this question could easily have been tweaked a little to be done using a calculator rather than R.

COVID and, more recently, COVID-19 vaccines are hot topics these days. Consequently, I wanted to know how we (i.e., Ontario) are doing regarding vaccination. I thus downloaded vaccination data from the [Ontario Data Catalogue](#). I could not get all the data I was initially looking for. However, I was able to get data on the number of first and second doses administered by date, age group, and [Public Health Units \(PHU\)](#) for the period from Jul 27 to Aug 10, 2021. The data I actually downloaded can be found [here](#). However, since that original dataset only provided the cumulative number of doses administered (by date, age group and PHU), I had to make a few simple calculations to get the number of doses administered each day. Hence, you should download this updated dataset from [STAT 332 LEARN website](#) (see file `vaxCovid19.csv` in the Exam folder) instead of the original dataset from the Ontario Data Data Catalogue. This file can be easily read into R by using the command `read.csv`, and it contains the following variables:

Variable	Description
<code>date</code>	Date (Jul 27 to Aug 10, 2021; thus a total of 15 days)
<code>PHU</code>	Name of the Public Health Units (there is a total of 34 PHU in Ontario)
<code>ageGrp</code>	age group (12–17, 18–29, 30–39, 40–49, 50–59, 60–69, 70–79 and 80+)
<code>doses</code>	Total number of doses administered that day (see below)
<code>population</code>	Population (see below)

For clarity:

- file `vaxCovid19.csv` contains the data for 15 days, 8 age groups and 34 PHU; consequently, it consists of a grand total of $15 \times 8 \times 34 = 4080$ observations
- `doses` is the total number of doses (both first and second doses) administered that day by that PHU and for that given age group
- `population` is the number of individuals for that given age group that resides in that PHU; it is based on Census data, and thus (obviously) does not change by date
- `doses` is the outcome; i.e., `doses` is y_{ijk}
- `date` and `ageGrp` are the 2 factors
- PHU is not a factor, but corresponds to the k index; hence, $n_0 = 34$ and thus $k = 1, \dots, 34$

- (a) Produce the two-way Anova table, and perform the usual F tests. What are your conclusions on the effects of age, date and their interaction?
- (b) What age group (or groups) was (were) getting the most vaccines during that period from Jul 27 to Aug 10? Does this make sense/is that what you expected?
- (c) What age group (or groups) was (were) getting the fewest vaccines during that period from Jul 27 to Aug 10? Does this make sense/is that what you expected?
- (d) Use a contrast to test (at the 5% level) if there was a weekend/holiday effect in the number of people getting vaccinated? In other words, are fewer/more vaccines administered on weekends/holidays than on weekdays? *Note that Aug. 2 was a civic holiday in Ontario.*
- (e) One important issue with the analysis conducted above in parts (a)–(d) is that it looks at the total number of doses administered, and thus fails to account for the size of the population. For example, **doses** for the PHU of Toronto are much higher than those of other PHU simply because it has a much larger population. Though not as dramatic, this also holds true for the different age groups. Hence, it is much better to conduct the analysis based on the number of doses as a percentage of the population than on **doses** itself. This is exactly why the dataset contains the **population** variable. To this end, let y_{ijk}^* be the the number of doses as a percentage of the population for the k^{th} PSU on the i^{th} day and for the j^{th} age group.
 - i) Fit another two-way Anova table using y_{ijk}^* is the outcome variable, and perform the usual F tests. What are your conclusions on the effects of age, date and their interaction? Are they the same as in part (a)?
 - ii) Does your conclusions to parts (b) and (c) remain the same?